The size of an orbit with a local crossing

Human Compression project: L044 continuation

All marked generators below are involutions.

Lemma 1 (An anchored action)**.** *Let $T = \langle t_0, \dots, t_{h-1} \rangle$ be transitive on V, where the t_i are involutions with string commutations, $h \geq 1$, and $t_{h-1} \neq 1$. Suppose $a \in V$ is fixed by $t_1, \dots, t_{h-1}$. Then $|V| \geq h + 1$. In equality $T = \mathrm{Sym}(V)$.*

Proof. For $h = 1$ this is the two-point orbit of a nontrivial involution. For $h \geq 2$, choose a shortest path from a, using $t_0, \dots, t_{h-2}$, to a point moved by t_{h-1}. Such a point exists, or the prefix orbit would be invariant under T and the last mark would fix all of V. Every preceding point on this path is fixed by the last mark, so its final edge adds a new point. The complete acquired no-gap argument makes every label $0, \dots, h-2$ occur: at a missing index, commute the higher factors, which fix a, past the lower factors, which commute with the last mark. Thus the path, including its final t_{h-1}-edge, has at least h edges, proving the bound. In equality its labels occur exactly once in increasing order. The complete acquired exhausted anchored-path lemma gives the natural symmetric group when $h \geq 3$; for $h = 2$ the two distinct transpositions on three points generate S_3. No independence or IP is required. □

Lemma 2 (Two commuting anchored factors)**.** *Suppose L, R commute and LR is transitive on A. At $a \in A$, the nonidentity left restrictions have string indices ending at $j - 1$, all except that last mark fixing a; the right restrictions start at $j + 1$, all except that first mark fixing a. Let h, u be their respective numbers of active labels. Then*

$$|A| \geq h + u + 1,$$

except possibly when $|A| = 2$, $h = u = 1$, and both factors induce the same transposition on A. An empty factor is permitted and has zero labels.

Proof. If A is a singleton, both factors are trivial and the bound holds. Hence assume $|A| \geq 2$. The active labels within each factor form one interval adjoining j. Indeed any other marked component fixes a and commutes with the remaining components, so is trivial on the transitive orbit of its own factor. Commutation with the opposite factor then makes it trivial on all of A. Any nonidentity restriction in a factor is also nonidentity on its orbit through a: if it fixed that orbit pointwise, commuting with the opposite factor would make it trivial on A. The preceding lemma, in reversed order on the left, gives $|a^L| \geq h + 1$ and $|a^R| \geq u + 1$ for nonempty factors.

If neither factor is transitive, its orbits have equal sizes: the opposite commuting factor permutes them transitively. Hence

$$|A| \geq 2 \max\{|a^L|, |a^R|\} \geq h + u + 2.$$

If L is transitive, every right mark other than the one of index $j + 1$ centralizes L and fixes a, so is trivial. Thus $u \leq 1$. For $u = 0$ the anchored bound suffices. For $u = 1$ the centralizer of L on A is nontrivial. If $h \geq 2$, equality $|A| = h + 1$ would make L natural symmetric of degree at least three, whose centralizer is trivial. Therefore $|A| \geq h + 2 = h + u + 1$. If $h = 1$, transitivity of its single involution forces $|A| = 2$, giving precisely the stated exception. The case with R transitive is the same with the factors interchanged. □

Theorem 3 (The whole local orbit)**.** *Let $H = \langle \rho_m, \dots, \rho_x \rangle$ act transitively on O, with string commutations and every displayed restriction nonidentity. Suppose the full local omission of some $j \in [m, x]$ has exactly two orbits A, B, joined by a unique j-edge $\{a, b\}$. Assume also that the omission of the first mark ρ_m is intransitive on O. Then*

$$|O| \geq x - m + 2.$$

Proof. Put $L = H_{<j}$ and $R = H_{>j}$. They commute and LR is transitive on each of A, B. Every distant mark fixes a, b: it preserves the local omission orbits and the unique crossing edge. Thus the two preceding anchored factors apply on each side. Write their active lengths as h_A, u_A and h_B, u_B, taking a zero for an empty factor. The active intervals adjoining j proved above give

$$\max(h_A, h_B) = j - m =: h, \qquad \max(u_A, u_B) = x - j =: u.$$

If these maxima are attained on different sides, the individual anchored bounds give $|A| + |B| \geq (h + 1) + (u + 1)$. If they are both attained on one side, say A, the preceding lemma gives $|A| \geq h + u + 1$ and the nonempty other side adds one more point.

The only unaccounted possibility is $h = u = 1$, $|A| = 2$ and $|B| = 1$. Then the first and last marks are the same transposition on A, both fixing B. The middle mark is the crossing transposition on the three points. Omitting the first mark leaves a transitive action, contrary to the hypothesis. If $|B| \geq 2$, the desired bound already holds. This proves all cases, including an empty left or right factor. □

Theorem 4 (When the central involution exchanges the witness orbit)**.** *Retain every actual odd-degree head hypothesis of* Local pairs and ambient witnesses*, including two ambient eligible edges globally for every tail colour. Let O be its normalized orbit with a genuine local unique crossing, with active interval $\{5, \dots, x\}$. If δ exchanges O with a distinct H-orbit, then*

$$x < r - 1, \qquad |\operatorname{Fix}(G_{>x})| \geq 2x + 1.$$

Proof. The first omission on O is $K = G_{>5}$ and is intransitive by normalization. The preceding theorem gives $|O| \geq x - 3$. Since δ centralizes H, its other orbit O' has the same complete marked action and the same size. All marks after x fix both orbits. They also fix the seven points $D \cup \{b\}$, disjoint from them. Therefore

$$|\operatorname{Fix}(G_{>x})| \geq 2(x - 3) + 7 = 2x + 1.$$

Suppose $x = r - 1$. The acquired actual anchor $z = b^{\rho_4}$ is fixed by δ and by K. It is outside D, b and is not H-fixed: otherwise the full group would preserve $D \cup \{b, z\}$, as in the complete last-fixed-point proof. It therefore belongs to a nontrivial H-orbit V disjoint from O, O' and the seven points. The equation $n = 2r + 1$ and $|O| = |O'| \geq r - 4$ leave at most two points for V. Consequently $|O| = r - 4$, $|V| = 2$, and these sets exhaust Ω. The group K fixes V pointwise, since it fixes its point z. Transitivity of $H|_V$ forces ρ_5 to exchange the two points of V. Its restrictions on O, O' are conjugate by δ and therefore have the same sign. It fixes D, b. Thus ρ_5 is odd on all of Ω, a contradiction. The terminal case is excluded. □

Theorem 5 (The witness orbit is fixed pointwise)**.** *Under the preceding actual head hypotheses, δ fixes O pointwise.*

Proof. First suppose δ exchanges O, O'. The preceding proof supplies $x < r - 1$ and the point $z = b^{\rho_4}$, which is fixed by δ, K but not by H. Thus $y = z^{\rho_5} \neq z$. Both lie in the same H-orbit, disjoint from O, O' and $D \cup \{b\}$.

If $x \geq 6$, every mark after x commutes with ρ_5 and fixes z, so fixes y as well. Together with O, O' and the seven head points, these give at least $2(x - 3) + 9 = 2x + 3$ fixed points. Apply the complete ambient interval bound to $I = \{x + 1, \dots, r - 1\}$. It is nonempty, every colour has

two ambient eligible edges, and its action is intransitive because of these fixed points. If U is its support and $c \geq 1$ its number of nontrivial orbits, that bound gives

$$|U| \geq 2(r-1-x)+c \geq 2(r-1-x)+1.$$

The disjoint fixed points and support then force $n \geq 2r+2$, contrary to $n = 2r+1$.

It remains that $x = 5$. The transitive action on O is a single nontrivial involution, so $|O| = |O'| = 2$. Since $x < r-1$, the interval $K = G_{>5}$ is nonempty. It fixes the twelve points

$$D \cup \{b\} \cup O \cup O' \cup \{z\}.$$

The same interval bound gives $f + c \leq 13$, where f counts its fixed points and $c \geq 1$ its nontrivial orbits. Therefore $f = 12$, $c = 1$; its unique nontrivial orbit U has odd size $n - 12 = 2r - 11$. The involution δ centralizes K, so preserves U. A nonidentity centralizer element of a transitive action fixes no point; an involution with no fixed point has even support. Since $|U|$ is odd, $\delta|_U = 1$. On the twelve fixed points it fixes D, b, z and exchanges O with O', acting as exactly two transpositions. It is therefore even everywhere, contrary to its stated oddness. This excludes all orbit exchanges.

Hence δ preserves O. The actual $\tau \in G_j$ induces δ there and centralizes its full H-action. It preserves the unique ambient witness edge, and ambient eligibility prevents it from exchanging that edge's endpoints. It fixes a point of a transitive H-orbit and centralizes H, so fixes the entire orbit pointwise. □

Attribution and remaining route. The anchored no-gap and exhausted-path proofs and full ambient interval bound are acquired. The two commuting-factor size count, terminal parity argument and twelve-fixed-point parity argument are the new steps. They avoid asserting disjointness of two point paths. The actual witness orbit and its normalized first omission come from the complete preceding proof. The remaining route is now the orbit fixed pointwise by δ, together with full source routing and global attachment. No new induction, computation, formal, original or question credit is claimed.